

• Problem Statement ID – 26127

• Problem Statement Title -

VeriTrack — City-Wide Multi-

Camera ANPR & Vehicle

Trajectory Analytics

• Theme - Transportation & Logistics

• PS Category - Software

• Team ID -

• Team Name - ASSASIN

THE CHALLENGE

Existing systems work at the level of individual cameras, but connecting the same vehicle across multiple cameras in a city remains difficult due to:

**Blurred / Partial Plates**

Plates are often unclear, occluded, or captured at angles, leading to OCR errors and inconsistent readings.

**False Vehicle Matches**

Similar-looking vehicles (same model/color) can cause mismatches, leading to incorrect associations.

**Indian Plate Variations**

Wide variations in number plate fonts, sizes, spacing, and formats across states make OCR inconsistent.

**Large Number of Camera Feeds**

Handling high volumes of video streams from distributed cameras requires significant processing and storage.

**Privacy & Surveillance Risk**

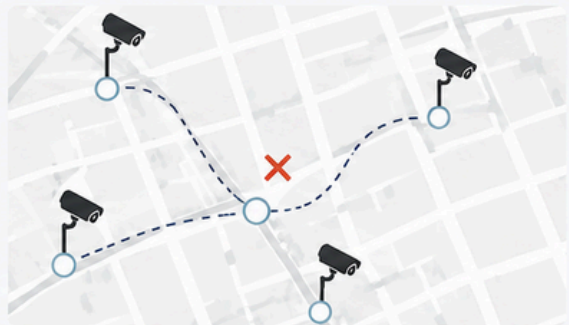
Tracking vehicles across locations raises privacy concerns and requires secure data handling and retention policies.

**Incorrect / Untrusted Alerts**

OCR or matching errors can trigger wrong alerts, reducing trust and increasing manual verification effort.

WHY IT MATTERS

Without reliable cross-camera association, complete vehicle journeys, route anomalies and city-wide movement patterns are difficult to reconstruct.



**Incomplete Journeys**

**Route Anomalies**

**Unreliable City-wide Insights**

Gurugram to test 380 ANPR traffic surveillance cameras in 3 weeks

GMDA and district police are implementing the surveillance network on Golf Course Road, MG Road and other high-density stretches.

Published on: May 10, 2026, 03:58:12 IST

By [HT Correspondent](#)



VERI-TRACK INTELLIGENT SURVEILLANCE

Multi-Camera Vehicle Intelligence & Traffic Command Platform

Command Center

Vehicle Search

Trajectories

Traffic Analytics

Alerts

Camera Network

System Health

EVALUATION TARGETS:

MH12AB1234 CRITICAL BLACKLIST

OD02XX1234 UNUSUAL DEVIATION

KA05MB4321 HIGH SPEED

DL01CA9988 CLONED SPATIAL CLASH

MH01DE5544 COMMERCIAL CORRIDOR

RECONSTRUCTED VEHICLE

MH12AB1234 (White Hyundai Creta)

TOTAL DISTANCE

18.6 km

TRANSIT DURATION

37.1 mins

AVG CORRIDOR SPEED

30.1 km/h

IDB

94

SPATIAL GIS TRAJECTORY PATH

Observed Sequence (5 Waypoints)

Play Timeline

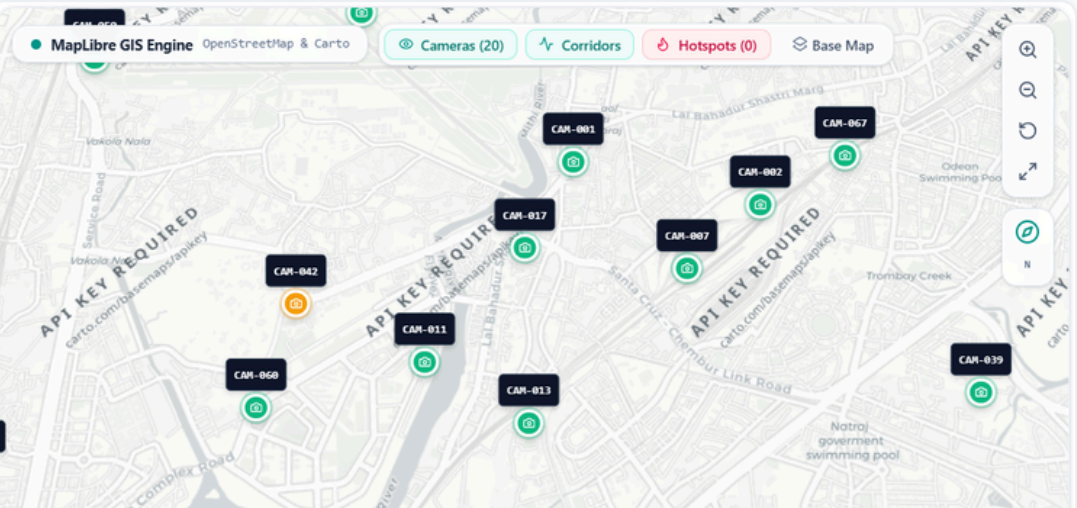
MapLibre GIS Engine OpenStreetMap & Carto

Cameras (20)

Corridors

Hotspots (0)

Base Map



CHRONOLOGICAL CAMERA OBSERVATIONS

1 CAM-001

Junction A (MG Road Intersect)

MH12AB1234 Direction: Southbound

Observed Transit Segment

2 CAM-007

Ring Road (Cloverleaf Exit)

MH12AB1234 Direction: Eastbound

Observed Transit Segment

3 CAM-013

City Center South Flyover Ramp

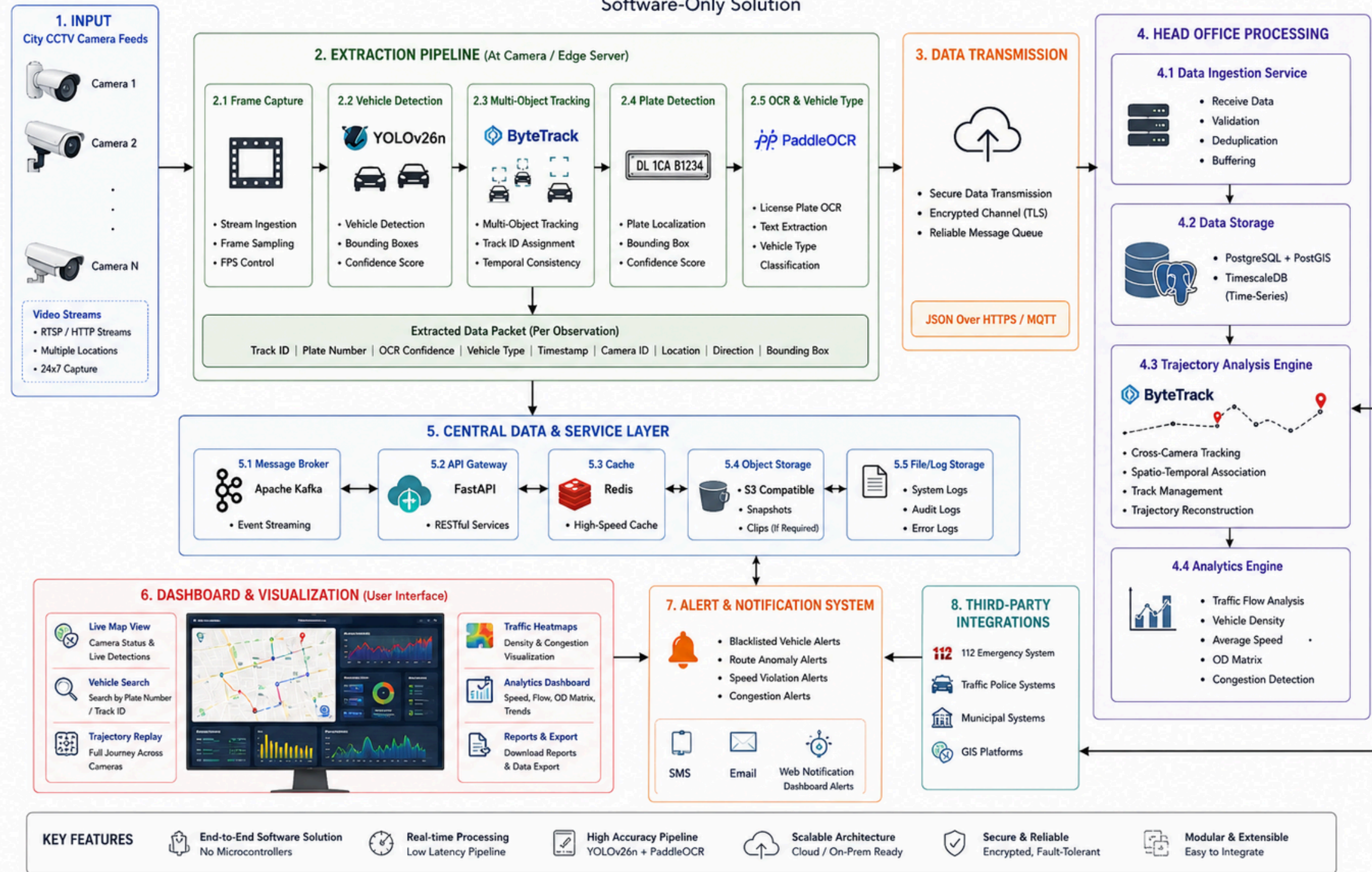
MH12AB1234 Direction: Southbound

@SIH 2026 Idea Submission - Team ASSASIN

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CITY-WIDE ANPR & TRAJECTORY ANALYTICS SYSTEM ARCHITECTURE

Software-Only Solution



Key Features

- India-Tuned Detection & Rectified OCR:** High-precision dual-head detection with 4-point perspective de-skewing and per-character confidence scoring, customized for all Indian plate types (HSRP, two-wheeler slimline, commercial yellow, EV green, and Bharat Series).
- Hybrid Trajectory Reconstruction:** Fuses license plate text, visual vehicle appearance (Re-ID), travel direction, and transit time into a unified confidence score to reconstruct continuous, map-matched vehicle journeys from full or partial plate queries.
- Macro Traffic Dynamics & O-D Analytics:** Generates real-time camera-to-camera Origin-Destination (O-D) flow matrices, live corridor speed tracking, and intersection congestion heatmaps.

Showstoppers & Differentiators

- Privacy-by-Architecture (DPDP Act Ready):** Protects citizen privacy with AES-256 encryption, tamper-proof audit trails, and 30-day evidence retention purges.
- Non-Disruptive Retrofit & 112 Dispatch:** Ingests existing CCTV camera feeds without pole or camera replacements, pushing instant alerts to dispatch systems.
- Occlusion Resilience & Transparent Scoring:** Tracks vehicles with damaged, or obscured plates using visual metric embeddings, providing sub-score breakdowns (OCR match, visual Re-ID, and transit feasibility) to eliminate false positives.
- Interactive Geospatial Command Console:** Interactive dashboard featuring real-time incident markers, dynamic traffic density heatmaps, and animated route timeline replays.

Developing a privacy-aware, India-tuned AI framework for city-scale traffic monitoring that fuses multi-camera ANPR, vehicle Re-ID embeddings, and spatio-temporal road network constraints for real-time trajectory reconstruction and urban traffic analytics.

Core Technical Highlights

Edge Detection & Tracking

- **YOLO26n (NMS-Free)**: 2.4M params, 6.8 ms latency (INT8), 94.8% mAP50.
- **ByteTrack**: Single-camera tracklet linking; eliminates frame duplicate processing.
- **4-Point Homography**: Rectifies angled plate crops before character recognition.

High-Accuracy Multi-Frame OCR

- **PP-OCRv6 Tiny**: 1.8 ms latency, >98.5% character accuracy, <1.8% error rate.
- **Multi-Frame Voting**: Aggregates character confidences across vehicle tracklet.
- **India-Tuned Validation**: Regex filters for HSRP, BH-Series, and 2-wheelers.

Visual Re-ID & Cross-Camera Matching

- **OSNet**: 128-D L2 vehicle embedding, 88.4% Rank-1 accuracy (VeRi-776).
- **Multi-Modal Fusion**: Fuses OCR + Visual Re-ID + Heading + Travel Time.
- **Adaptive Fallback**: Automatically shifts weight to Re-ID on muddy/broken plates.

System Speed & Statutory Governance

- **End-to-End Latency**: ~11.8 ms per vehicle pass (Orange Pi 5 / RK3588).
- **Stream Throughput**: ~110 FPS (Processes 3x 1080p @ 30 FPS streams).
- **Emergency 112 Dispatch**: Sub-2s automated blacklist alert push.
- **DPDP Act Compliance**: AES-256 field encryption with 30-day auto-TTL purge from storage.

1. INPUT

Multi-Camera Video Feeds

- RTSP Streams
- Intersections
- Highways
- Toll Plazas
- Parking Areas
- ...

Frame Sampling
(Adaptive FPS)

Key Frame Extractor

2. PREPROCESSING

- Bilinear Resize (640×640×3 Tensor)
 $I_r \in \mathbb{R}^{640 \times 640 \times 3}$
- CLAHE Contrast (clipLimit = 3.0, 8×8 Grid)
 $I_c = \text{CLAHE}(I_r)$
- Bilateral Denoising ($d = 9, \sigma = 30$)
 $I_d = \text{Bilateral}(I_c)$
- Laplacian Focus Gating ($\text{Var}(\nabla^2 I_d) \geq 75$ & Aspect Ratio 2.0–6.0)
Focus Score = $\text{Var}(\nabla^2 I_d)$

Normalized Frames

3. DETECTION

- 3A. Vehicle Detection (YOLO26n)
NMS-Free (Anchor-Free)
- YOLO26n Spec:
 - 2.4M Parameters
 - 5.4 GFLOPs (640×640)
 - 6.8 ms (INT8, CPU)
 - mAP@0.5: 82.6% (COCO)

- 3B. License Plate Detection (YOLO26n – LP Head)
→ Plate Bounding Boxes

ABC 1234

Plate Crop & Rectification

- Perspective Transform
- Skew Correction
- Size Normalization

DL 1CA B1234

4. OCR – CORE

PP-OCRv6 Tiny Rec
(SVTR-LCNet + CTC Decoder | 1.8 ms INT8)

Detection (DBNet) → Recog. Head (SVTR-LCNet) → CTC Decoder → Post-Processing (Regex + Rules)

Outputs

Raw Text: D1CAB1234
Corrected Text: DL1CAB1234

India HSRP Regex (State Code + Pattern):

 $\wedge\{A-Z\}\{2\}\{0-9\}\{1,2\}\{A-Z\}\{1,2\}\{0-9\}\{4\}\$$

State Code Dictionary (All India)

Character Accuracy > 98.5% | CER < 1.8%

Per-Character Confidence (C_{char}):

D	L	1	C	A	B	1	2	3	4
0.98	0.97	0.99	0.95	0.96	0.98	0.98	0.97	0.96	0.98

Full String Confidence (C_{ocr}): 0.962

4B. Plate Format Validation (India-Tuned)

- State Code Dictionary
- Format Rules (e.g., XX00XX0000)
- HSRP / Color / Series Validation

Final Plate Text

DL1CAB1234

 $C_{ocr} = 0.962$

5. VEHICLE RE-ID (VISUAL)

OSNet Omni-Scale Backbone
(256×128 Crop Input | 3.5 ms INT8)

128-D L2-Normalized
Feature Embedding
($\|f_v\|_2 = 1$)
 $f_v \in \mathbb{R}^{128}$

Benchmark:
88.4% Rank-1 Accuracy
on VeRi-7

6. HYBRID MATCHING ENGINE (ACROSS CAMERAS)

For a candidate pair ($i \leftrightarrow j$)

Matching Factors:

- AB Text Similarity (Levenshtein)
- V Visual Similarity (OSNet Cosine)
- θ Heading Angle Consistency
- v Speed Gate ($v = d/\Delta t$)

Hybrid Score:

$$S = w_T \cdot S_{text} + w_V \cdot S_{visual} + w_\theta S_\theta + w_v S_v$$

where $\sum w_k = 1$

 $S \geq 0.75$?

NO

Low Confidence Match
(Queue for Review)

YES

Match Confirmed
(Link Established)

8. MACRO TRAFFIC ANALYTICS

O-D Flow Matrix
($T_{OD} \in \mathbb{R}^{|O| \times |D|}$)

$$T_{OD} = \begin{bmatrix} t_{11} & \dots & t_{1D} \\ \vdots & \ddots & \vdots \\ t_{O1} & \dots & t_{OD} \end{bmatrix}$$

Link Volume & Spatial Density Heatmaps

Corridor Speed Profiles & Bottleneck Forecasting

Analytics Outputs

- Traffic Flow & Density (Temporal + Spatial)
- Origin-Destination Patterns
- Dwell Time Heatmaps
- Anomaly Detection (Isolation Forest / Autoencoder)
- Corridor Performance Index (CPI)
- Bottleneck & Congestion Forecasting

Data Source: Re-ID Tracks + Plate Reads + Telemetry
Storage: TimescaleDB + PostGIS

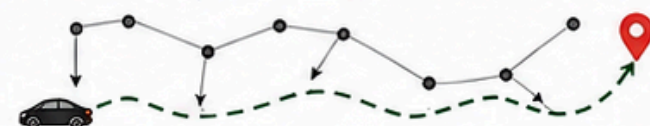
7. TRAJECTORY RECONSTRUCTION – CORE

Sighting Tuple at Time t_i
 $p_i = (Cam_i, Lat_i, Lon_i, t_i, \theta_i)$

where:

- Cam_i : Camera ID
- (Lat_i, Lon_i) : GPS / Geo-coordinates
- t_i : Timestamp
- θ_i : Heading Angle

HMM Map-Matching to Road Graph $G = (V, E)$
(OSRM / PostGIS)

Path Inference
(Viterbi on HMM States)Emissions: $p(p_i | state_k)$ Transitions: $p(state_k \rightarrow state_{k+1})$

Kinematic Speed Check

 $v = d_{road} / \Delta t$ Flags $v > 150$ km/h
as Cloned Plate

Outputs: Trajectory ID | Path (Node Sequence) | Dwell Time | Avg Speed | Stops | Entry/Exit Points

Feasibility

Challenge

Solution

Technical Feasibility (System Requirements):

- **Edge Resource Budget:** Complete AI pipeline (YOLO26n + PP-OCRv6 + ByteTrack + OSNet) executes in under 12 ms per vehicle pass within an 850 MB RAM and 12W power footprint on low-cost edge hardware.
- **Real-Time Multi-Stream Processing:** Handles continuous 1080p @ 30 FPS video feeds with zero frame drops using hardware VPU decoding and adaptive 5 to 10 FPS decimation (~110 FPS stream caapcity).
- **High-Accuracy Recognition:** Exceeds mandatory 90% plate accuracy under adverse conditions (night IR, 45-degree pole angles, motion blur) via 4-point homography de-skewing and multi-frame voting.
- **Sub-2s Alert Latency:** Delivers sub-1.8-second hotlist alert latency from initial camera passage to emergency dispatch triggering.

Operational Feasibility (Field Deployment):

- **Non-Disruptive Retrofit:** Directly connects to existing municipal ONVIF and RTSP CCTV camera feeds with zero pole replacements or road digging required.
- **Automated Law Enforcement Dispatch:** Sub-2-second automated hotlist matching pushes verified alerts, GPS coordinates, and vehicle crops directly to National Emergency 112 (CAD) and e-Challan systems.
- **Explainable Operator Scoring:** Multi-signal confidence verification (OCR text score, visual appearance match, and road-transit speed check) eliminates false alarms and operator fatigue.

Economic Feasibility (Cost & ROI):

- **75% Lower Edge Hardware CapEx:** Deploys on low-cost edge gateways at approximately 110 dollars per node, compared to 500+ dollars for industrial GPU servers.
- **Zero Recurring Software Licensing Fees:** 100% open-source software stack eliminates proprietary VMS licensing costs (saving 12,000 to 25,000 rupees per camera annually).
- **99.4% Network Bandwidth Reduction:** Local edge inference transmits only 3.3 KB structured JSON telemetry per pass, reducing 500-camera network demand from 2.5 Gbps to under 18 Mbps.

Blurred /
Partial Plates

- 4-point homography perspective de-skewing unwarps angled plates into flat canonical 128x48 patches before recognition.
- Temporal multi-frame character voting aggregates predictions across 8 to 12 frames per tracklet to eliminate motion blur.
- OSNet visual appearance embeddings maintain vehicle identity when plates are physically unreadable.

False Vehicle
Matches

- Spatio-temporal road-graph speed gating ($v = d / dt$) validates physical travel feasibility, flagging impossible jumps.
- Multi-modal weighted fusion combines Levenshtein text distance, OSNet cosine similarity, and travel-time physics.
- Low-confidence matches are routed to server-side heavy VLM fallback rather than triggering false alarms.

Indian Plate
Variations

- India-tuned PP-OCRv6 Tiny Rec trained on HSRP, BH-Series, two-wheeler slimline, commercial yellow, and EV green plates.
- Deterministic regex engine validates state codes, RTO district numbering, and character checksums.
- Handles non-standard fonts and regional spacing anomalies without requiring separate models.

Large Number of
Camera Feeds

- Edge-decoupled computing executes YOLO26n and PP-OCR locally at the roadside on low-cost gateways (Orange Pi 5 / RK3588).
- Transmits only 3.3 KB structured JSON telemetry per vehicle pass, cutting network bandwidth by 99.4% (2.5 Gbps down to 18 Mbps for 500 cameras).
- Central Kafka event bus ingests telemetry into zone-partitioned topics without central GPU bottlenecks.

Privacy &
Surveillance Risk

- DPDP Act 2023 compliance via AES-256 field-level encryption on plate records and strict role-based access control (RBAC).
- Tamper-proof cryptographic audit logs track every investigator query.
- Automated 30-day Time-To-Live (TTL) storage policy permanently purges raw plate crop evidence.

Untrusted /
False Alerts

- Multi-signal alert thresholding requires concurrent confirmation (OCR confidence >85%, visual Re-ID match, and road feasibility) before triggering Emergency 112 dispatch.
- Cloned plate anomaly engine automatically flags duplicate plates active across disparate city sectors at impossible speeds (>150 km/h).
- Explainable alert scoring displays individual sub-score breakdowns (OCR, Re-ID, speed) on the SOC console.

Potential Impact

- **Faster detection** of suspicious vehicles across multiple camera locations in real time.
- **Improved road** safety through early identification of wrong-way driving, congestion, and abnormal vehicle activity.
- **Enhanced law-enforcement capabilities** by reconstructing vehicle movement across geographically distributed cameras.
- **Better traffic management** through real-time traffic density, travel-time, and origin–destination insights.
- **Scalable smart-city infrastructure** that can be extended across highways, intersections, toll plazas, and parking areas.

Benefits

- **End-to-end automated monitoring** — from video capture and detection to matching, tracking, and alerts.
- **Higher identification accuracy** through combined OCR, visual Re-ID, direction, and travel-time analysis.
- **Reduced manual surveillance effort**, allowing authorities to focus on verified alerts and critical incidents.
- **Real-time actionable intelligence** through dashboards, GIS visualization, and automated notifications.
- **Modular and cost-effective deployment** using open-source AI technologies and scalable cloud-native architecture.

Existing Systems:

01	 YOLOv26n (Ultralytics YOLO) <i>Object Detection</i>	Key Strengths - Real-time and lightweight object detection. - Optimized for speed and accuracy. - Supports export (ONNX, TensorRT) for deployment.	Why Relevant / Reference Official Repository: GitHub – Ultralytics YOLO https://github.com/ultralytics/ultralytics Official Documentation: Ultralytics YOLO Docs https://docs.ultralytics.com/
02	 OSNet (Vehicle Re-Identification) <i>Visual Re-ID</i>	Key Strengths - Omni-Scale feature learning for robust embeddings. - Strong generalization across cameras & conditions. - Generates compact 128-D / 512-D embeddings.	Why Relevant / Reference Official Repository (Torchreid): KaiyangZhou/deep-person-reid https://github.com/KaiyangZhou/deep-person-reid Research Paper: Omni-Scale Feature Learning for Re-Identification https://arxiv.org/abs/1905.00953 Pretrained Weights: Hugging Face – kaiyangzhou/osnet https://huggingface.co/kaiyangzhou/osnet
03	 ByteTrack (Multi-Object Tracking) <i>Vehicle Tracking</i>	Key Strengths - Real-time multi-object tracking. - Associates high & low score detections. - Handles occlusion and crowded scenes.	Why Relevant / Reference Official Repository: GitHub – FoundationVision/ByteTrack https://github.com/FoundationVision/ByteTrack Research Paper: ByteTrack: Multi-Object Tracking by Associating Every Detection Box (ECCV 2022) https://arxiv.org/abs/2110.06864
04	 UP Police – Trinetra Platform (AI-CCTV System) <i>Large-Scale Deployment</i>	Key Strengths - AI-powered criminal & vehicle tracking platform. - Integrated with 350,000+ CCTV feeds. - Real-time alerts, face & vehicle analytics.	Why Relevant / Reference IndiaAI Official Case Study – Trinetra https://indiaai.gov.in/case-study/ai-is-being-used-by-the-up-police-to-catch-criminals News & Deployment Record: Times of India – Trinetra 2.0 https://timesofindia.indiatimes.com/city/lucknow/up-police-launches-trinetra-20-running-on-crime-gpt/articleshow/108520268.cms
05	 Surat Smart City ITMS & Safe City (Traffic Management) <i>Smart City Deployment</i>	Key Strengths 2,000+ CCTV grid with video analytics. - Traffic management, e-Challan & congestion control. - Integrated command & control system.	Why Relevant / Reference NIUA Case Study – Surat Safe City & Video Analytics https://iccc.niua.org/ Smart Cities Mission – MoHUA Case Studies https://smartcities.gov.in/
06	 IHMCL – ANPR Standards (Indian Highways) <i>ANPR Infrastructure</i>	Key Strengths - Defines ANPR deployment standards for India. - Covers Multi-Lane Free Flow (MLFF) environments. - Ensures interoperability & reliability.	Why Relevant / Reference Official Guidelines: IHMCL – Electronic Toll & ANPR Standards https://ihmcl.co.in/